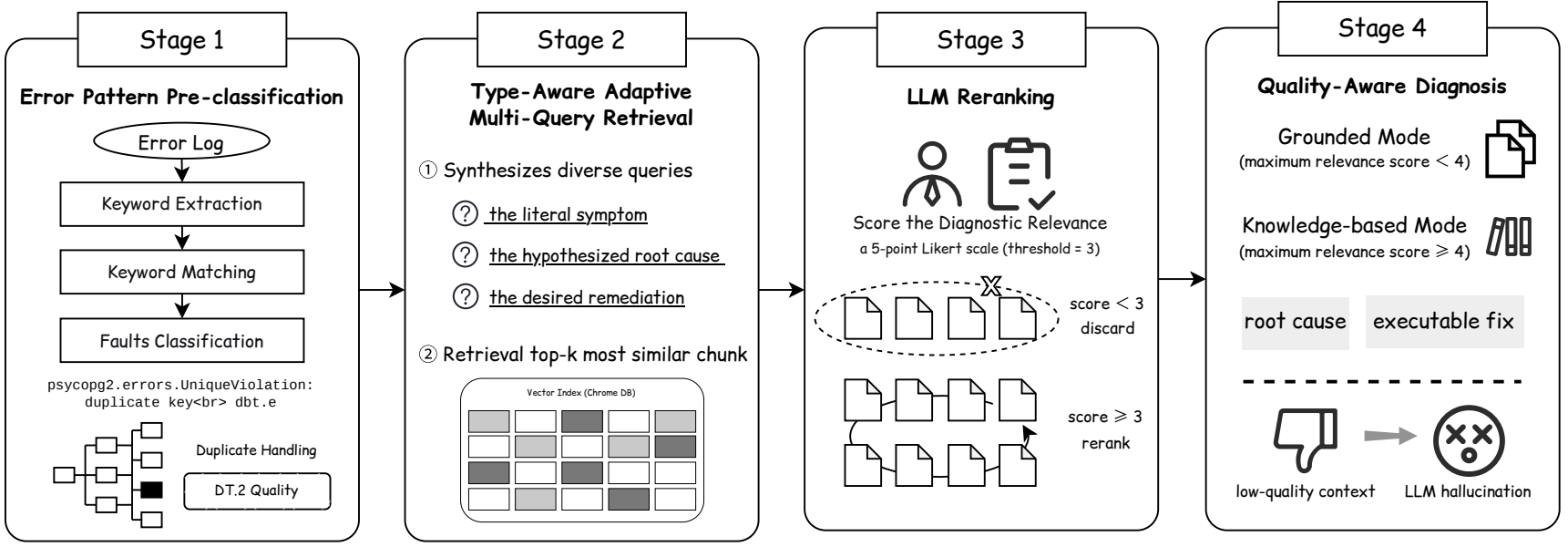

Stage 1
Error Pattern Pre-classification
Error Log
Keyword Extraction
Keyword Matching
Faults Classification
psycopg2.errors.UniqueViolation:
duplicate key
 dbt.e
Duplicate Handling
DT.2 Quality
Stage 2
Type-Aware Adaptive
Multi-Query Retrieval
① Synthesizes diverse queries
the literal symptom
the hypothesized root cause
the desired remediation
② Retrieval top-k most similar chunk
Vector Index (Chrome DB)
Stage 3
LLM Reranking
Score the Diagnostic Relevance
a 5-point Likert scale (threshold = 3)
score < 3
discard
score ⩾ 3
rerank
Stage 4
Quality-Aware Diagnosis
Grounded Mode
(maximum relevance score < 4)
Knowledge-based Mode
(maximum relevance score ⩾ 4)
root cause
executable fix
low-quality context
LLM hallucination